K-Means Clustering Algorithm | Geometric Intuition

1. What is K-Means?

- **K-Means** is an **unsupervised learning** algorithm.
- It groups data into **K clusters** based on similarity.
- Each cluster is represented by its **centroid (mean point)**.
- Goal: Minimize the distance between data points and their cluster centroid.

👉 Think of it as: "Find groups in the data where each group is centered around a mean point."

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2. Step-by-Step Intuition

➢ ***Understanding K-Means with a Problem Statement***

Suppose you run a store and want to group customers based on their spending habits. Instead of labeling them manually, K-Means finds natural groupings:

- High spenders
- Medium spenders
- Low spenders

➢ Deciding the Number of Clusters (K)

- K is chosen before training.
- Too small K → groups are too broad.
- Too large K → groups may be meaningless.
- Best way: Use **Elbow Method** (explained later).

➢ Initializing Centroids

- Start by choosing **K random points** from the dataset as "centroids" (temporary centers of clusters).

➢ Assigning Clusters

- For each data point:
    1. Calculate distance (usually Euclidean) to each centroid.
    2. Assign the point to the **nearest centroid** → forms clusters.

➢ Updating Centroids

- After assignment, compute the **mean position** of points in each cluster.
- Move the centroid to this mean.

- Repeat **assign → update** steps until centroids stop moving (convergence).

👉 This ensures clusters are compact and well-separated.

🤔 Should We Stop Clustering?

- Yes, when:
    1. Centroids stop changing significantly.
    2. Or max iterations are reached.

➢ K-Means Clustering (Elbow Method)

- To choose the **best K**, run K-Means with different K values.
- Plot **WCSS (Within-Cluster Sum of Squares)** vs. K.
- WCSS = how spread out the points are within clusters.
- The curve usually decreases quickly at first, then slows down.
- The "elbow point" = best K (balance between too few and too many clusters).

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3. Key Points to Remember

- **Unsupervised** → no labels required.
- Sensitive to **initial centroids** (different starts can give different clusters).
- Works best when clusters are **spherical and equally sized**.
- Variants like **K-Means++** improve initialization.

✓ Summary:

K-Means groups data into K clusters by iteratively assigning points to the nearest centroid and updating centroids until stable. The **Elbow Method** helps decide the right K.
